Rohit Kumar

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SUMMARY

B.Tech. CSE (AI Major) student at IIITDM Kancheepuram passionate about building scalable software systems and intelligent AI solutions. Experienced in full-stack engineering, deep learning, and high-performance computing with hands-on projects in ML, computer vision, and distributed systems.

EDUCATION

Indian Institute of Information Technology, Kancheepuram *Aug 2023 – 2027*
B.Tech. in Computer Science and Engineering *(Major: Artificial Intelligence)* CGPA: **9.14 / 10**

TECHNICAL SKILLS

Languages:	C++, JavaScript, Python, C, SQL
Frontend:	React.js, Next.js, Tailwind CSS, Redux Toolkit, HTML5
Backend:	Node.js, Express.js, REST, JWT, FastAPI
Databases / Cloud:	MongoDB, MySQL, AWS (S3)
AI-ML:	PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas, OpenCV
High Performance Computing:	CUDA, OpenMP, MPI
Tools:	Git, GitHub, Postman, Vercel, Render

PROJECTS

Bone Age Prediction from Hand Radiographs — *Python, PyTorch, scikit-learn, OpenCV, NumPy*

Comparative study of Classical ML vs. Deep Learning for automated pediatric bone age assessment using the RSNA dataset (12,611 X-rays).

- Trained a **multi-task ResNet34** with demographic (sex) embeddings for simultaneous age regression and developmental stage classification, achieving **1.646-year MAE** and **79.3% accuracy** — a 25.3% improvement over the classical baseline.
- Implemented a Classical ML pipeline: HOG feature extraction (8,100-dim vectors) with Ridge Regression for regression and Random Forest for classification (67.1% accuracy, MAE 2.205 years).

Text-Guided Brain Tumor Segmentation using Vision-Language Models — *Python, PyTorch*

Multimodal deep learning framework jointly leveraging FLAIR MRI and radiology reports for pixel-wise tumor segmentation via a text-conditioned VLM architecture.

- Designed **TG-SwinUNet** on the **BraTS 2020** dataset integrating a Swin Transformer encoder with `BiomedVLP-CXR-BERT`; introduced multi-level **FiLM fusion** at three skip-connection levels and a **cross-attention bottleneck** (8 heads, image Q / text K,V) for spatial word-to-region mapping.
- Implemented **Text-Guided Attention Gates (TGAG)** and **Deep Supervision** heads with a compound loss (Dice + Focal + Boundary + CLIP-style Contrastive), achieving **Dice 0.856**, **IoU 0.776**, and **Hausdorff 21.66 px** on BraTS 2020 validation set.
- Proved text modality benefit via strict ablation over an identical image-only baseline (**+0.010 Dice, +0.011 IoU, −1.32 px HD**); validated cross-modal alignment with t-SNE and model focus with Grad-CAM.

AttendX — *React.js, Express.js, MongoDB, HTML5 QR Code API, JWT* [Live] [GitHub]

Smart QR-based attendance management system with role-based access and real-time GPS validation.

- Rolled out to **200+ students and 2 faculty**; logged 500+ secure scans, reducing manual record-keeping effort by **80%**.
- Secured APIs with **JWT + HTTP-only cookies**; tokens auto-expire after configurable idle periods to mitigate replay attacks.

ACHIEVEMENTS

- **GATE CSE 2026** — secured AIR **3966** among thousands of candidates nationwide.
- **Semifinalist, Flipkart GRiD 7.0** — advanced to the semifinal round in a national-level coding competition.
- **LeetCode Knight** — achieved a peak contest rating of **1904**; solved **500+ problems** across LeetCode & Codeforces.

RELEVANT COURSEWORK

Data Structures & Algorithms — Object-Oriented Programming — Relational Database Management System — Computer Networks — Operating Systems — Pattern Recognition & Machine Learning — Deep Learning — Reinforcement Learning — Applied Data Science — Artificial Intelligence — High Performance Computing — Digital Image Processing — Internet of Things